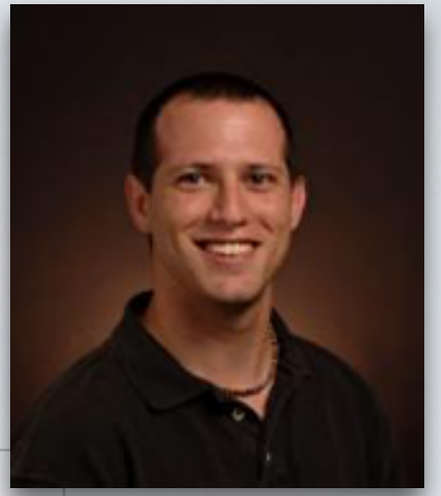
DOING BAYESIAN DATA ANALYSIS ? OR NOT ?

#6 Introduction to bayesian statistical modeling 28.06.2016

OVERVIEW



Context	Normal Distribution Models	Model Comparisons
Machinery of Probability	Markov Chain Monte Carlo	Factor Analysis
Interpretations of Probability	Regression	Structural Equation Modeling
Review of Frequentist Inference	Multilevel Modeling	Item Response Theory
Introducing Bayesian Inference	Specifying Priors	Latent Class Models
Beta-Bernoulli & Beta-Binomial	Posterior Predictive Distributions	Bayesian Networks
Summarising posteriors	Missing Data	Summary & Additional Resources
Introduction to Graphical Models	Posterior Predictive Model Checking	
Accumulation of Evidence		

"The theory of inverse probability [referring to Bayes' inference] is founded upon an error, and must be wholly rejected."

— Fisher (1958)

KOLMOGOROV'S AXIOMS

- Start with a single variable X with a finite set of mutually exclusive and exhaustive possible values $(x_1,..,x_J)$
- Probability mass functions (pmf) specify the probability for each state

$$\text{Non-negativity}: 0 \leq (x_j) \leq 1$$

$$\text{Normalization}: \sum_{j=1}^{J} p(x_j) = 1$$

$$\text{Finite additivity}: P(X = x_j \text{ or } X = x_m) = p(x_j) + p(x_m)$$

Kolmogorov, A.N. (1933). "Foundations of the Theory of Probability".

JOINT DISTRIBUTION

	x_1	x_2
y_1	.1	.2
y_2	.3	.1
y_3	.2	.1

Joint probability distribution of X and Y

$p(x_1, y_1) = .1$

$p(x_1, y_2) = .3$

$p(x_2, y_3) = .1$

$$\sum_{j=1}^{J}\sum_{k=1}^{K} p(x_j, y_k) = 1$$

MARGINAL DISTRIBUTION

- The **marginal** probability of X taking on a value x_j is the sum over all possible joint probabilities involving that value of X

$$p(x_j) = \sum_{k=1}^{K} p(x_j, y_k)$$

- This can be viewed as a marginalization over Y

MARGINAL DISTRIBUTION

	x_1	x_2
y_1	.1	.2
y_2	.3	.1
y_3	.2	.1
	.6	.4

Marginal probability of X

$$p(x_1) = p(x_1, y_1) + p(x_1, y_2) + p(x_1, y_3)$$
$$= .1 + .3 + .2$$
$$= .6$$

$$p(x_2) = p(x_2, y_1) + p(x_2, y_2) + p(x_2, y_3)$$
$$= .2 + .1 + .1$$
$$= .4$$

CONDITIONAL DISTRIBUTION

- Discrete random variables X and Y

- $P(X = x_j \mid Y = y_k) = p(x_j \mid y_k)$ is the probability that X takes on the value x_j **given that** Y takes on the value y_k

$$p(X = x_j \mid Y = y_k) = \frac{P(X = x_j, Y = y_k)}{P(Y = y_k)}$$

CONDITIONAL DISTRIBUTION

	x_1	x_2
y_1	.1	.2
y_2	.3	.1
y_3	.2	.1
	.6	.4

Probabilities for Y conditional on $X = x_1$

$$p(y_1 \mid x_1) = p(x_1, y_1) \,/\, p(x_1)$$
$$= .1 \,/\, .6$$
$$= .17$$

$$p(y_2 \mid x_1) = p(x_1, y_2) \,/\, p(x_1)$$
$$= .3 \,/\, .6$$
$$= .60$$

CONNECTING JOINT, MARGINAL AND CONDITIONAL

- The joint probability of a particular value of X and a particular value of Y is the product of the conditional probability for X given that value of Y and the marginal probability for that value of Y

$$p(x_j, y_k) = p(x_j \mid y_k) p(y_k)$$

- The marginal probability of X is the sum of its conditional probabilities given all possible values of Y, where each conditional probability is weighted by the associated marginal probability for that value of Y

$$p(x_j) = \sum_{k=1}^{K} p(x_j \mid y_k) p(y_k)$$

DERIVING BAYES' THEOREM

$p(x, y) = p(x \mid y)\, p(y)$	The joint probability for x and y is the product of the conditional probability for x given y and the marginal probability for y
$p(x, y) = p(y \mid x)\, p(x)$	Reversing the roles of x and y
$p(x, y) = p(x, y)$	Tautology
$p(y \mid x)\, p(x) = p(x \mid y)\, p(y)$	Substitution
$p(y \mid x) = p(x \mid y)\, p(y) \,/\, p(x)$	Division
$p(x \mid y) = p(y \mid x)\, p(x) \,/\, p(y)$	Works both way

PROBABILITY FOR CONTINUOUS RANDOM VARIABLES

- Probability defined over ranges of values
- The full range is referred to as the **support**
  - The support for a normal (Gaussian) distribution is $(-\infty, \infty)$
- Probability density functions (pdf) rather than probability mass functions (pmf)
- For $x \sim N(\mu, \sigma^2)$: $$f(x) = \frac{1}{\sqrt{2\pi\sigma}} \exp\left[\frac{-1}{2\sigma^2}(x-\mu)^2\right]$$

PROBABILITY FOR CONTINUOUS VARIABLES

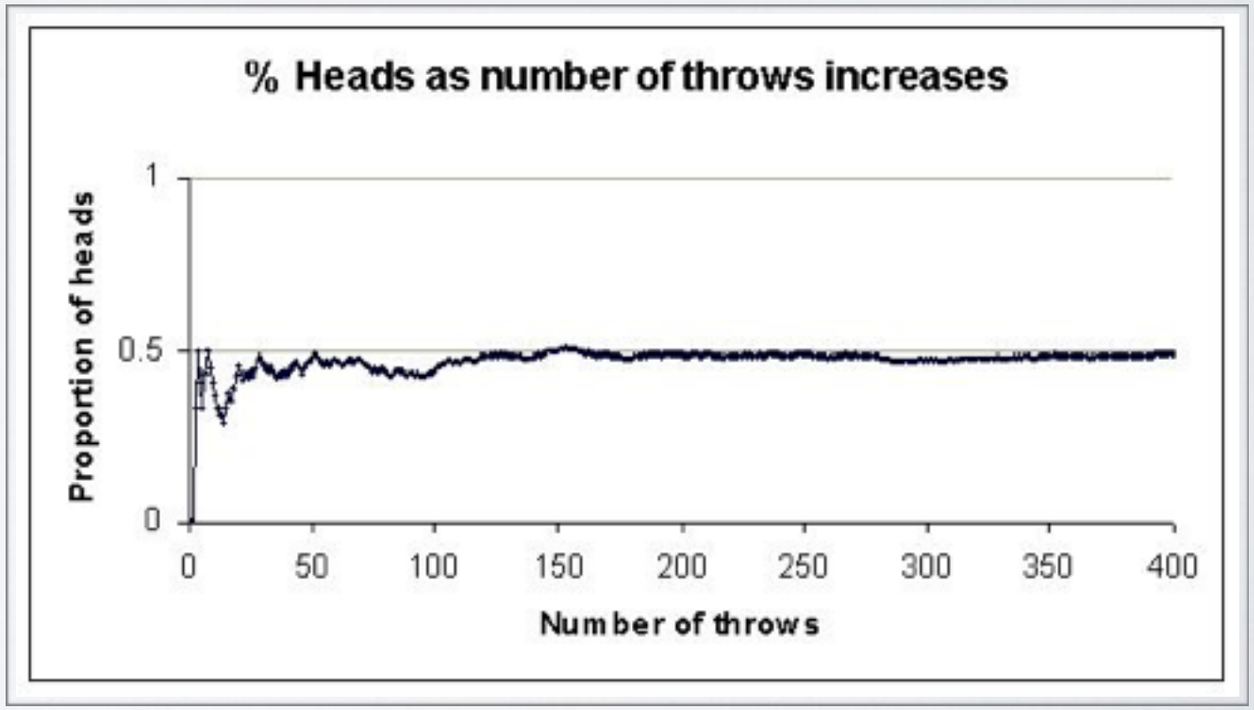
Discrete	Continuous
$\sum_{j=1}^{J} p(x_j) = 1$	$\int_x p(x) = 1$
$\sum_{j=1}^{J} \sum_{k=1}^{K} p(x_j, y_k) = 1$	$\int_x p(x,y) = 1$
$p(x_j) = \sum_{k=1}^{K} p(x_j, y_k)$	$p(x) = \int_y p(x,y)$
$p(x_j) = \sum_{k=1}^{K} p(x_j \mid y_k) p(y_k)$	$p(x) = \int_y p(x \mid y) p(y)$

Conceptions of probability

EMPIRICAL PROBABILITY (OR “FREQUENTIST”)

- Frequentist probability or *frequentism* defines an event's probability as the limit of its **relative frequency**
- Probability is a **property of the world**
- A claim of the frequentist approach is that in the "long run", the relative frequency will converge **exactly** to the true probability

$$P(x) = \lim_{nt \to \infty} \frac{n_x}{n_t}$$



SUBJECTIVE PROBABILITY ("SUBJECTIVE BAYESIANISM")

- Subjective probability is an agent's **degree of belief** that an event will occur

  What is the probability that the Dow Jones average will go up tomorrow ?

- P(x) = degree of belief that X is true

- "So to assign equal probabilities to two events is not in any way an assertion that they must occur equally often in any random experiment as Jeffreys emphasized it is only **a formal way of saying I don't know**." (Jaynes, 1985)

$$p(H \mid DI) = \frac{p(H \mid I)\, p(D \mid HI)}{p(D \mid I)}$$

$$S_I = -\sum p_i \log p_i$$

INTERPRETATIONS OF PROBABILITY

Characteristic	Frequentist	Subjective
Refers to	Physical events	Beliefs, uncertainty
Property of	The world	Your state of knowledge
Compatibility ?	Not with subjective	Yes
Inferential frame	Freq. & Bayes	Bayes
Applicability	Narrow	Broad

Review of frequentist inference

SET UP FOR INFERENCE

- There is an unknown parameter θ
- There is a **random variable** X defined on some sample space
- The sample data $x = (x_1, x_2, \ldots, x_n)$ are observations of the random variable, x is a **realisation** of the random variable X

INFERENCE

- There is an unknown parameter θ
    - Does **not** vary with sampling
    - **Not** a random variable, **not an object of probability**
- Want to use the sample data x to inform about an unknown θ
- Define an **estimator** as function of the random variable $\hat{\theta}(X)$
- The actual value it takes on, is called the **estimate**
- Example:
    - *θ is the population mean*
    - *$\hat{\theta}(X)$ is the mean in a sample (generically)*
    - *$\hat{\theta}(x)$ is the mean from the sample x*

SAMPLING DISTRIBUTION OF THE SAMPLE MEAN ESTIMATOR

$$p(\bar{X}) = N(\mu, \sigma^2_{\bar{X}})$$

$$p(\bar{X} > \mu) = .50$$

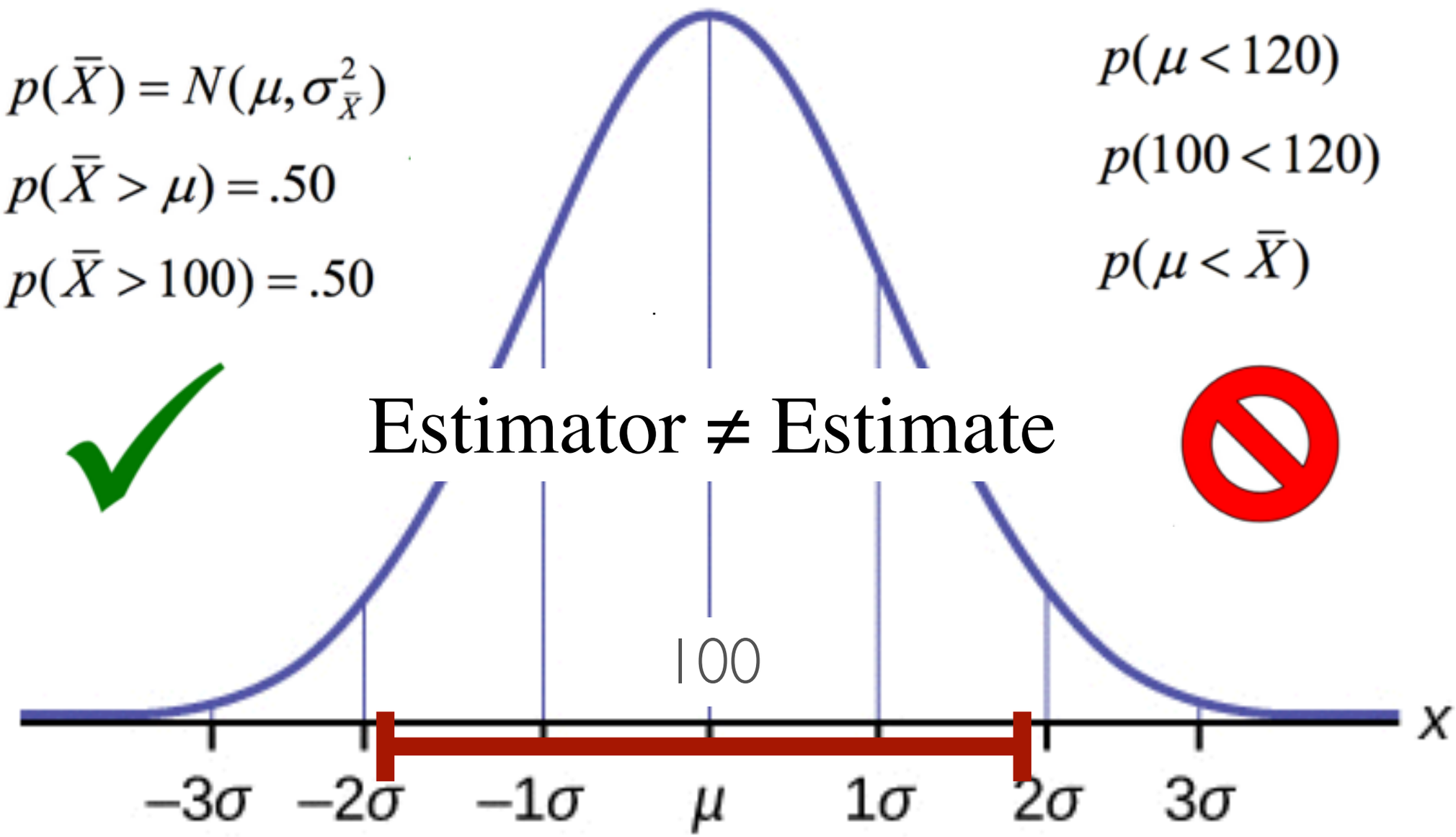
$$p(\bar{X} > 100) = .50$$

$$p(\mu < 120)$$

$$p(100 < 120)$$

$$p(\mu < \bar{X})$$

Estimator ≠ Estimate



CONFIDENCE INTERVALS FOR A POPULATION MEAN

- Probability interpretation is for the **estimator**

$$p(\hat{\theta}(X) < \theta) = 1 - \alpha$$

- Over repeated samples, the proportion of (X) less than θ will be $1 - \alpha$
- The proportion of times interval contains θ is $1 - \alpha$
- Do not know whether a particular (x) is less than θ
- Do not know whether a particular interval contains θ

http://rpsychologist.com/d3/CI/

Toward bayesian inference - example

EXAMPLE : MEDICAL DIAGNOSIS (GIGERENZER, 2002)

- Consider women aged 40-50 with no family history of cancer and no symptoms of cancer
    - The proportion that have breast cancer is .008
- Conduct mammogram screening
    - If a woman has breast cancer, the probability of a positive mammogram is .90
    - If a woman does not have breast cancer, the probability of a positive mammogram is .07
- A woman undergoes the mammogram, the result is positive. What's the probability the woman has breast cancer?

MAXIMUM LIKELIHOOD (ML)

- Maximum likelihood (ML) estimation asks: "What are the values of the parameters that make the data most probable?"
- Specify the conditional probability of the data as $p(x \mid \theta)$
    - E.g., $\theta = (\mu, \sigma^2)$ $p(x \mid \mu, \sigma^2) = N(\mu, \sigma^2)$
- When taken as a function of θ, this is referred to as the **likelihood**

$$L(\theta \mid x) = p(x \mid \theta)$$

- ML estimation then maximizes this function with regards to θ, using the known values of x

CONDITIONAL PROBABILITY DISTRIBUTION

- If a woman has breast cancer, the probability of a positive mammogram is .90

$$p(Mam = + \mid BC = Y) = .90, p(Mam = - \mid BC = Y) = .10$$

- If a woman does not have breast cancer, the probability of a positive mammogram is .07

$$p(Mam = + \mid BC = N) = .07, p(Mam = - \mid BC = N) = .93$$

	Mammogram result	
Breast cancer	Positive	Negative
Yes	.90	.10
No	.07	.93

CONDITIONAL PROBABILITY DISTRIBUTION

- A woman undergoes the mammogram, the result is positive

$$L(BC = Y \mid Mam = +) = p(Mam = + \mid BC = Y) = .90$$

$$L(BC = N \mid Mam = +) = p(Mam = + \mid BC = N) = .07$$

	Mammogram result	
Breast cancer	Positive	Negative
Yes	.90	[illegible]
No	.07	.93

- ML: *What is the value of BC that maximizes this? BC = Yes*

Bayesian inference

BAYES' THEOREM IN ACTION

- The setup:
  - Two entities : x (data) and θ (parameter)
  - We know the conditional probabilities $p(x \mid \theta)$, which tell us what to believe about x if we knew the value of θ
  - When we learn the value of x, what should we believe about θ

- We combine three things (two of which are important)
  - Relative conditional probabilities for x given θ, the likelihood
  - Previous probabilities about θ
  - The marginal probability for x

$$p(\theta \mid x) = \frac{p(x \mid \theta)p(\theta)}{p(x)}$$

THE PRIOR DISTRIBUTION

Prior = everything we know about θ before we observe value for x

Prior distribution for unknown θ

$$p(\theta \mid x) = \frac{p(x \mid \theta)\boxed{p(\theta)}}{p(x)}$$

The proportion of women aged 40-50 with no family history of cancer and no symptoms that have breast cancer is .008

CONDITIONAL PROBABILITY OR LIKELIHOOD

Conditional probability of x given θ; likelihood for θ

Prior distribution for unknown θ

$$p(\theta \mid x) = \frac{\boxed{p(x \mid \theta)}\,\boxed{p(\theta)}}{p(x)}$$

Relative probability of x given the value of θ

Once value of x is known, and this is viewed as a function of θ it is a **likelihood**

MARGINAL PROBABILITY

Conditional probability of x given θ; likelihood for θ

Prior distribution for unknown θ

$$p(\theta \mid x) = \frac{\boxed{p(x \mid \theta)}\boxed{p(\theta)}}{\boxed{p(x)}} \propto p(x \mid \theta)p(\theta)$$

Marginal probability of x (over θ)

$$p(x) = \sum_{\theta} p(x \mid \theta)p(\theta)$$

Serves to normalize the distribution. Note that $p(x)$ does not vary with θ. We will eventually discard it

MARGINAL PROBABILITY OF MAMMOGRAM = +

$$p(x) = \sum_{\theta} p(x \mid \theta) p(\theta)$$

$p(x \mid \theta)$

	Mammogram	
BC	Positive	Negative
Yes	.90	
No	.07	.93

X

×

$p(\theta)$

BC	
Yes	.008
No	.992

=

$p(x,\theta)$

BC	
Yes	.00720
No	.06944

Sum = .07664

POSTERIOR DISTRIBUTION

Conditional probability of x given θ; likelihood for θ

Prior distribution for unknown θ

$$\boxed{p(\theta \mid x)} = \frac{\boxed{p(x \mid \theta)}\boxed{p(\theta)}}{\boxed{p(x)}} \propto p(x \mid \theta)p(\theta)$$

Posterior probability distribution for unknown θ given x

Marginal probability of x (over θ)

Captures what we think about θ now that we have incorporated x

MEDICAL DIAGNOSIS POSTERIOR DIST.

$$p(\theta \mid x) = \frac{p(x \mid \theta)p(\theta)}{p(x)} = \frac{p(Mam = + \mid BC)p(BC)}{p(Mam = +)}$$

	Mammogram	
BC	Positive	Negative
Yes	.90	.10
No	.07	.93

X

×

BC	
Yes	.008
No	.992

.07664

The posterior probability that the woman has breast cancer

=

BC	
Yes	.09
No	.91

General approach to bayesian modeling

A ***Bayesianly justifiable*** analysis is one that

> *"treats known values as observed values of random variables, treats unknown values as unobserved random variables, and calculates the conditional distribution of unknowns given knowns and model specifications using Bayes' theorem."*

— Rubin (1984)

Beta-Bernoulli and Beta-Binomial Models

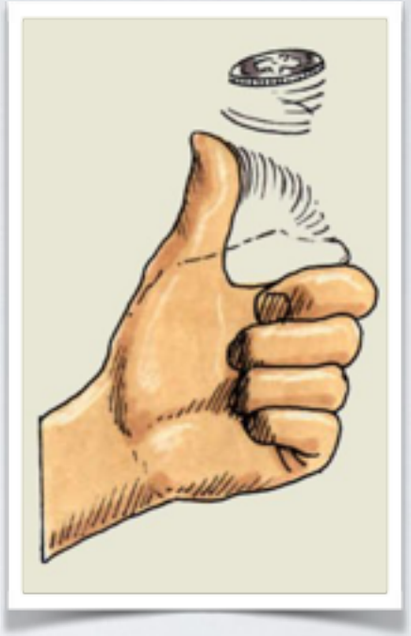
BERNOULLI DISTRIBUTION

- The probability that x takes on a value of 1 is governed by a parameter θ:

$$p(x = 1 \mid \theta) = \theta$$

- The (conditional) probability of the data is given by:

$$p(x \mid \theta) = \theta^{x} \times (1-\theta)^{(1-x)}$$



- Suppose we have J i.i.d. random variables $(x_1, x_2, \ldots, x_J)$:

$$p(x_1, \ldots, x_J \mid \theta) = \prod_{j=1}^{J} p(x_j \mid \theta) = \prod_{j=1}^{J} \theta^{x_j} (1-\theta)^{1-x_j}$$

- Example: $J = 2$ and $\theta = .7$

$$p(x_1 = \text{Face}, x_2 = \text{Face} \mid \theta) = \theta \times \theta = .7 \times .7 = .49$$

$$p(x_1 = \text{Face}, x_2 = \text{Pile} \mid \theta) = \theta \times (1\text{-}\theta) = .7 \times (1\text{-}.7) = .21$$

$$p(x_1 = \text{Pile}, x_2 = \text{Face} \mid \theta) = (1\text{-}\theta) \times \theta = (1\text{-}.7) \times .7 = .21$$

$$p(x_1 = \text{Pile}, x_2 = \text{Pile} \mid \theta) = (1\text{-}\theta) \times (1\text{-}\theta) = (1\text{-}.7) \times (1\text{-}.7) = .09$$

Binomial distribution

BINOMIAL DISTRIBUTION

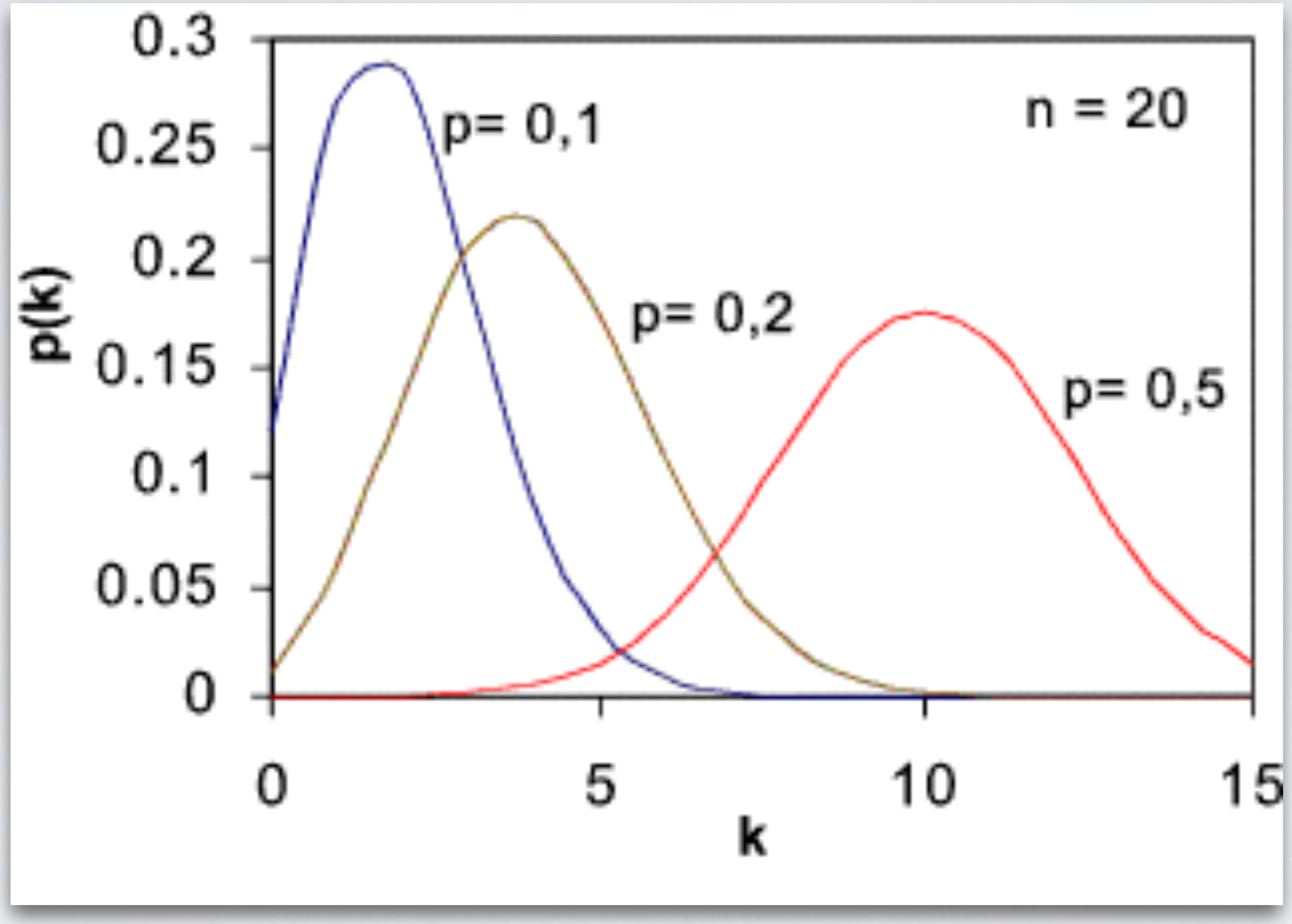
- Let $y = \Sigma x$ be the total number of successes in J Bernoulli trials each with probability θ
- The conditional probability of y, given θ, J, is

$$p(y \mid J, \theta) = Bin(y \mid J, \theta) = \begin{pmatrix} J \\ y \end{pmatrix} \theta^{y} (1-\theta)^{J-y}$$

θ raised to the number of successes

(1-θ) raised to the number of failures

BINOMIAL DISTRIBUTION



Introduction to WinBUGS

WINBUGS CODE FOR BERNOULLI AND BINOMIAL

```
model{
theta <- .7
x ~ dbern(theta)
y ~ bin(theta, J)
}
list(J=10)
```

$\theta = .7$

$x \sim \text{Bernoulli}(\theta)$

$y \sim \text{Binomial}(\theta, J)$

$J = 10$

Bernoulli and Binomial Distributions.odc

Bayesian inference

BACK TO BAYES

Conditional probability of x given θ; likelihood for θ

Prior distribution for unknown θ

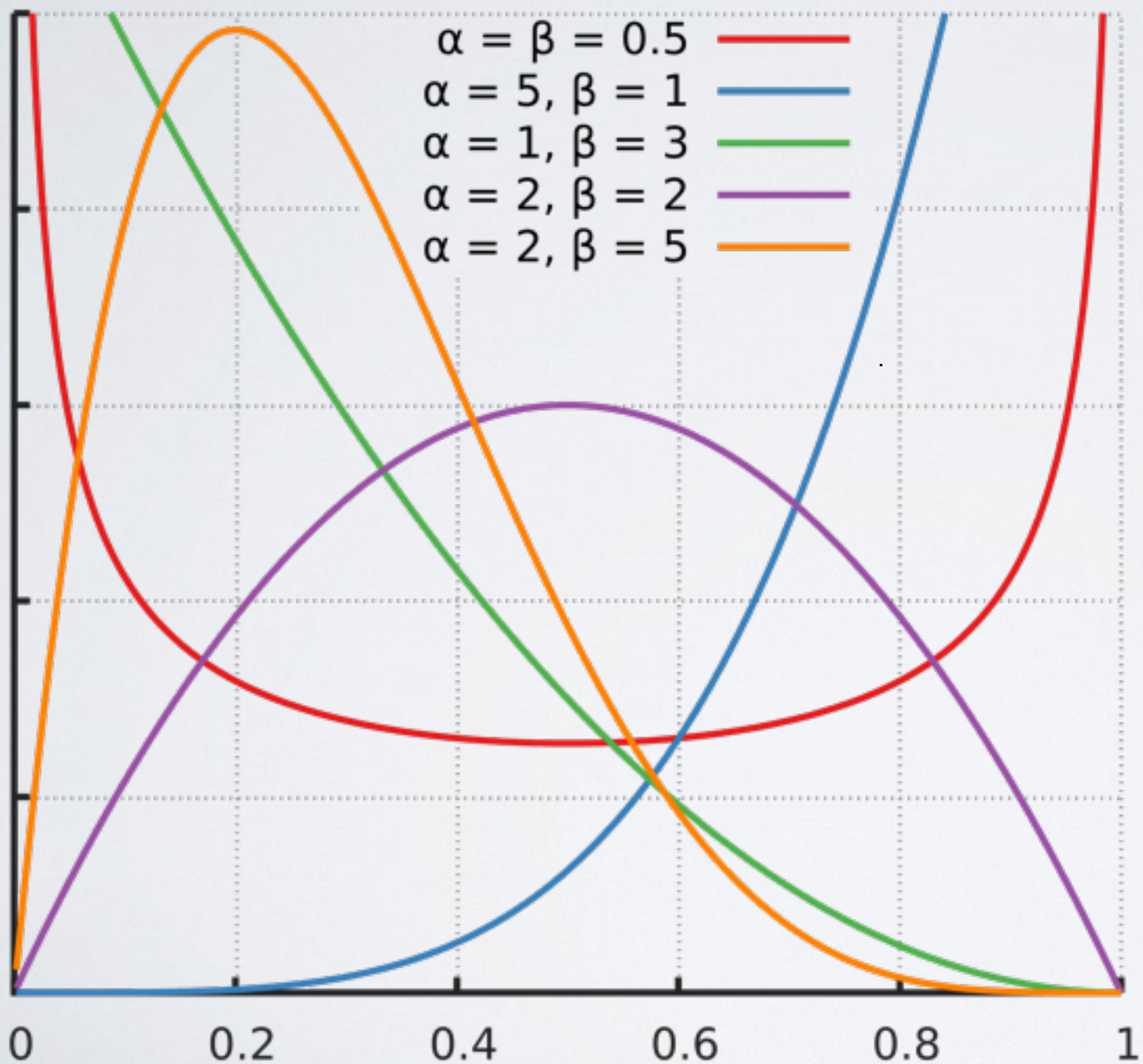
$$p(\theta \mid y, J) = \frac{\boxed{p(y \mid J, \theta)}\boxed{p(\theta)}}{p(y)}$$

A popular choice for a prior distribution in this context is a ***Beta*** distribution

$$p(y \mid J, \theta) = Bin(y \mid J, \theta) = \binom{J}{y} \theta^{y} (1-\theta)^{J-y}$$

Beta distribution

BETA DISTRIBUTION



- Distribution over [0,1]

$x \mid \alpha, \beta \sim \text{Beta}(\alpha, \beta)\ x^{\alpha-1}(1-x)^{\beta-1}$

$$E[Beta(\alpha,\beta)] = \frac{\alpha}{\alpha+\beta}$$

$$Var[Beta(\alpha,\beta)] = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

$$Mode[Beta(\alpha,\beta)] = \frac{\alpha-1}{\alpha+\beta)-2}$$

BACK TO BAYES

Prior distribution for unknown θ

$$p(\theta) = \text{Beta}(\alpha, \beta)\ \theta^{\alpha - 1}(1 - \theta)^{\beta - 1}$$

$$\boxed{p(\theta \mid y, J)} = \frac{\boxed{p(y \mid J, \theta)}\boxed{p(\theta)}}{\boxed{p(y)}} \propto p(y \mid J, \theta)p(\theta)$$

BACK TO BAYES

$$\boxed{p(\theta \mid y, J)} = \frac{\boxed{p(y \mid J, \theta)}\boxed{p(\theta)}}{\boxed{p(y)}} \boxed{\propto p(y \mid J, \theta) p(\theta)}$$

Marginal distribution of y (over θ)

In this case is a Beta-Binomial…

We will focus ignore and focus on proportionality

POSTERIOR DISTRIBUTION

$$p(\theta \mid y, J) \propto p(y \mid J, \theta)\, p(\theta)$$

$$\propto Bin(y \mid J, \theta)\, Beta(\alpha, \beta)$$

$$\propto \theta^{y}(1-\theta)^{J-y}\, \theta^{\alpha-1}(1-\theta)^{\beta-1}$$

θ raised to the number of successes

(1-θ) raised to the number of failures

POSTERIOR DISTRIBUTION

$$p(\theta \mid y, J) \propto p(y \mid J, \theta)\, p(\theta)$$

$$\propto Bin(y \mid J, \theta)\, Beta(\alpha, \beta)$$

$$\propto \theta^{y}(1-\theta)^{J-y}\, \theta^{\alpha-1}(1-\theta)^{\beta-1}$$

Suggests an interpretation of the prior as akin to $\alpha - 1$ successes and $\beta - 1$ failures

POSTERIOR DISTRIBUTION

$$p(\theta \mid y, J) \propto p(y \mid J, \theta)\, p(\theta)$$

$$\propto Bin(y \mid J, \theta)\, Beta(\alpha, \beta)$$

$$\propto \theta^{y}(1-\theta)^{J-y}\, \theta^{\alpha-1}(1-\theta)^{\beta-1}$$

$$\propto \theta^{y+\alpha-1}(1-\theta)^{J-y+\beta-1}$$

$$p(\theta \mid y, J) = Beta(y+\alpha, J-y+\beta)$$

The Beta is a ***conjugate prior*** for the binomial likelihood

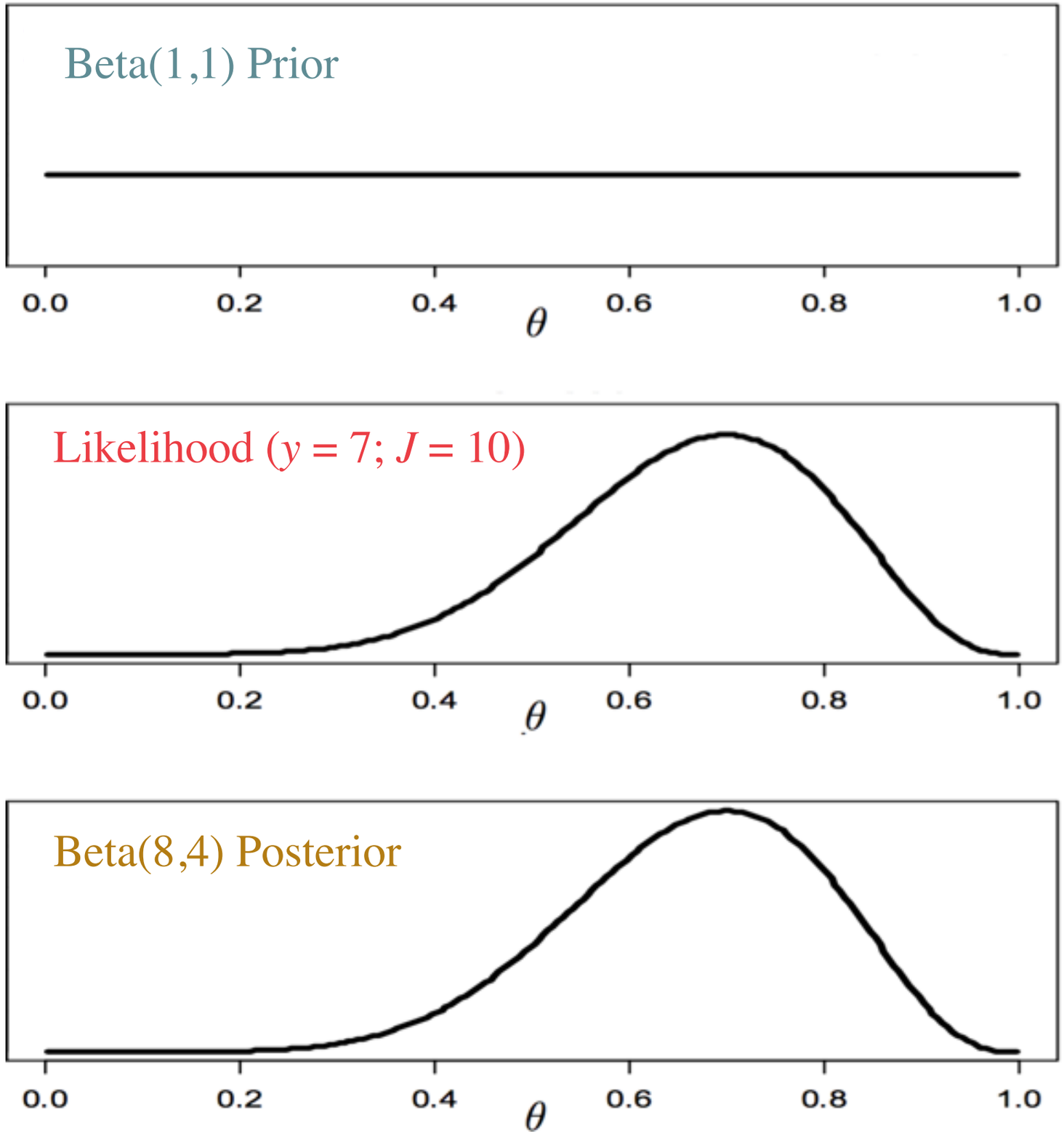
Example

EXAMPLE

- Observe y successes in J trials
    - E.g., 7 correct responses on 10 tasks
- Prior distribution as Beta(1,1)
    - Akin to 0 prior successes and 0 prior failures
- Posterior

$$p(\theta \mid y, J) \propto p(y \mid J, \theta)\, p(\theta) = \mathrm{Bin}(7 \mid 10, \theta)\, \mathrm{Beta}(1, 1)$$

$$p(\theta \mid y, J) = \mathrm{Beta}(y + \alpha, J - y + \beta) = \mathrm{Beta}(8, 4)$$


Beta(1,1) Prior
0.0
0.2
0.4
θ
0.6
0.8
1.0
Likelihood (y = 7; J = 10)
0.0
0.2
0.4
θ
0.6
0.8
1.0
Beta(8,4) Posterior
0.0
0.2
0.4
θ
0.6
0.8
1.0

POSTERIOR MEAN

$$p(\theta \mid y, J) \propto p(y \mid J, \theta)\, p(\theta) = Bin(7 \mid 10, \theta)\, Beta(1, 1)$$

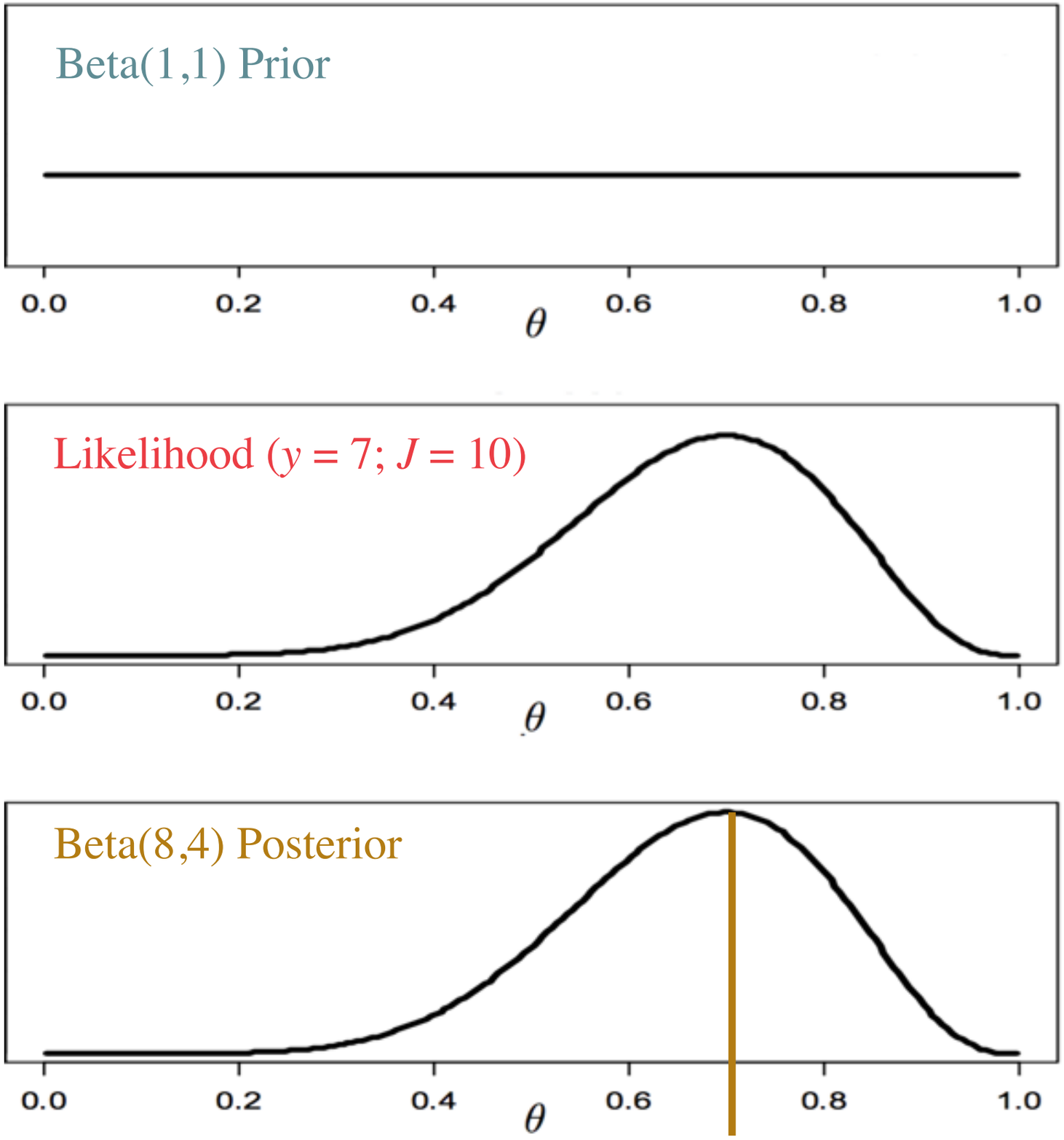
$$p(\theta \mid y, J) = Beta(y + \alpha, J - y + \beta) = Beta(8, 4)$$

$$\mu_{\theta|y,J} = \frac{y+\alpha}{y+\alpha+J-y+\beta} = \frac{y+\alpha}{\alpha+J+\beta} \boxed{= \frac{7+1}{1+10+1} \simeq .67}$$

$$= \left(\frac{J}{\alpha+\beta+J}\right)\left(\frac{y}{J}\right) + \left(\frac{\alpha+\beta}{\alpha+\beta+J}\right)\left(\frac{\alpha}{\alpha+\beta}\right)$$

Weighted average of mean in the data and mean in the prior

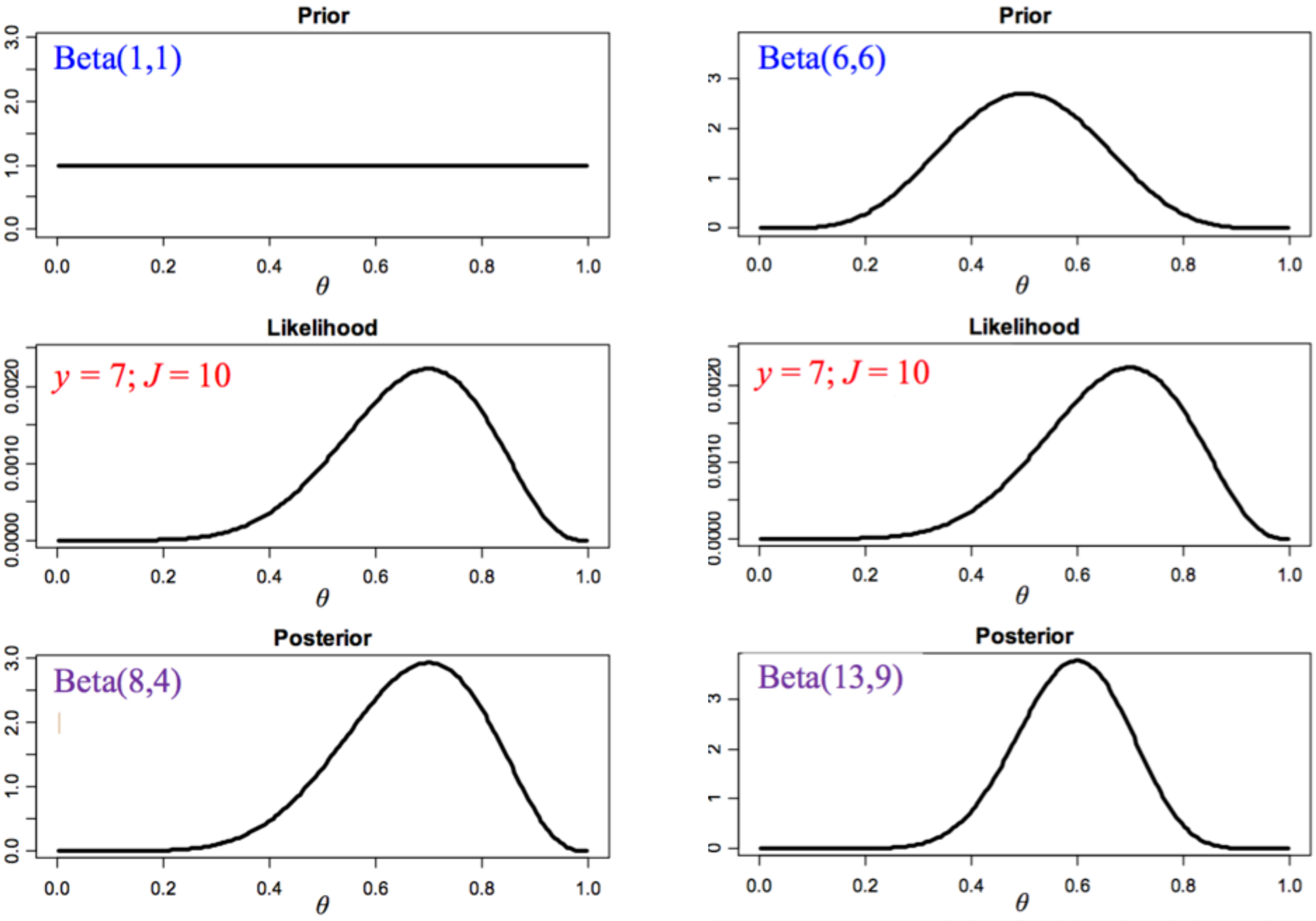
Weights are akin to the relative size of information in the data and relative size of the information in the prior

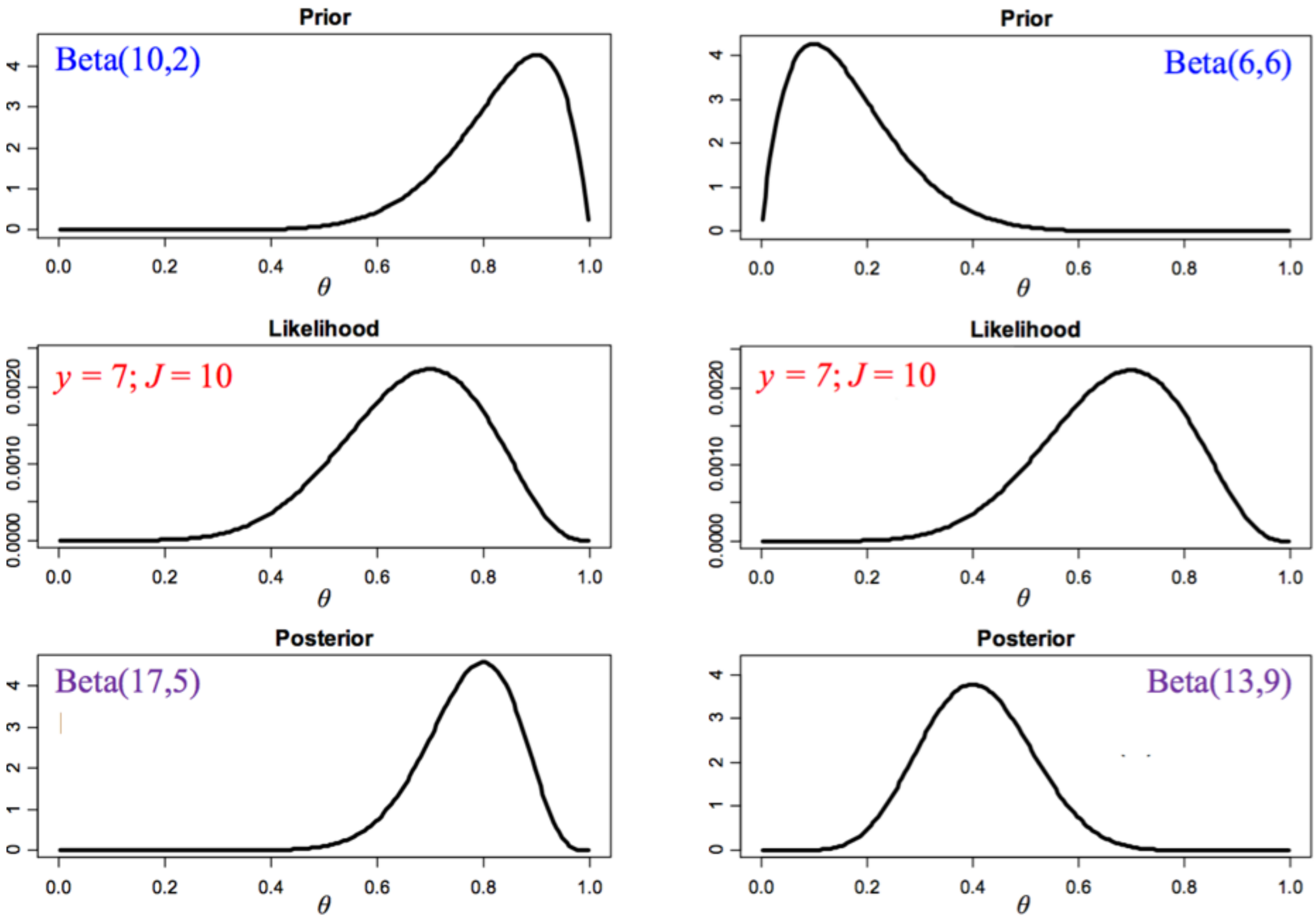


Posterior mean = 0.67

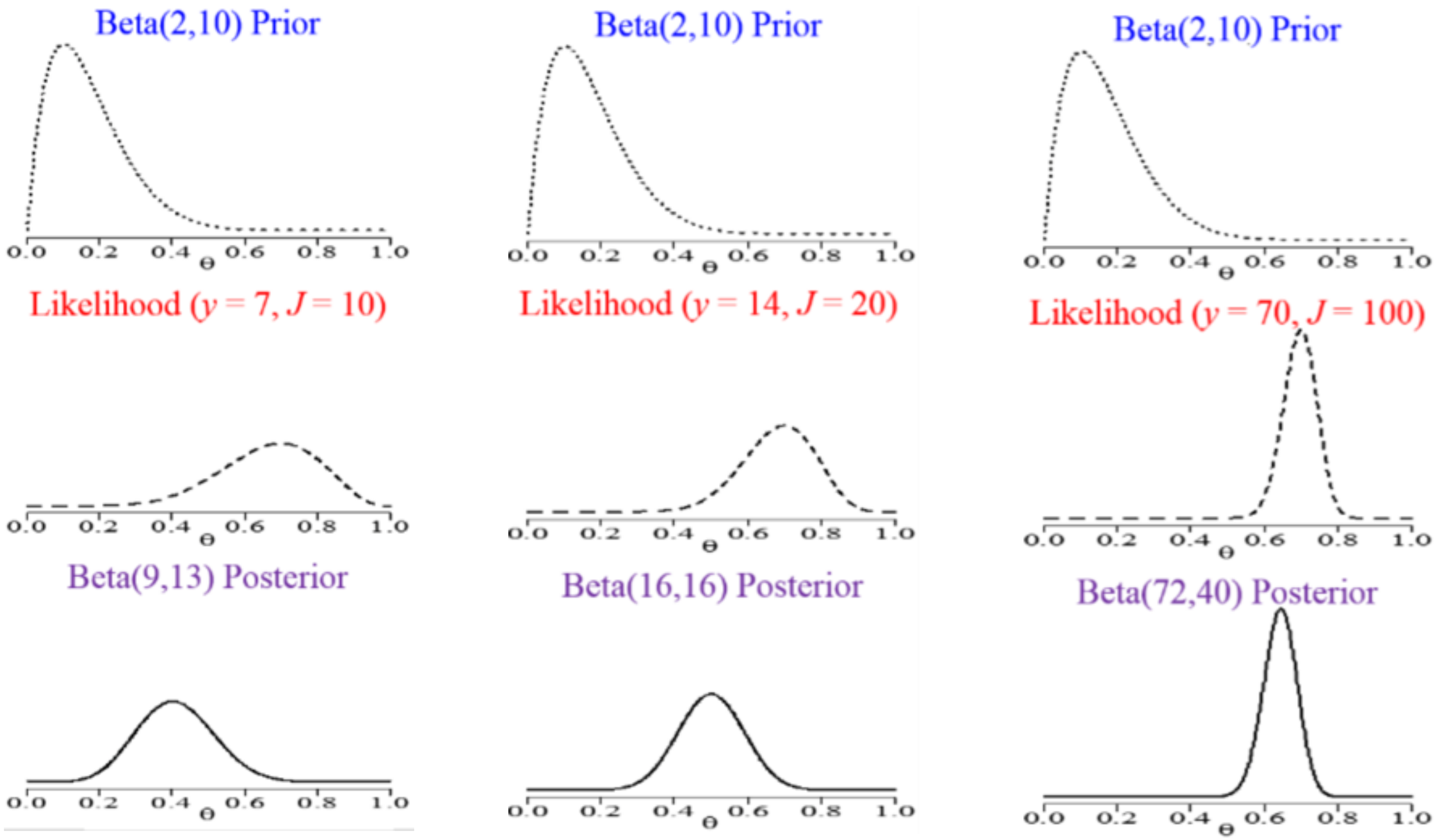
Posterior as synthesis of prior and likelihood

Effect of the information in the prior


Prior
Beta(1,1)
Prior
Beta(6,6)
Likelihood
$y = 7; J = 10$
Likelihood
$y = 7; J = 10$
Posterior
Beta(8,4)
Posterior
Beta(13,9)
θ


Prior
Beta(10,2)
Prior
Beta(6,6)
Likelihood
$y = 7; J = 10$
Likelihood
$y = 7; J = 10$
Posterior
Beta(17,5)
Posterior
Beta(13,9)
θ

Effect of the information in the data


Beta(2,10) Prior
Beta(2,10) Prior
Beta(2,10) Prior
0.0 0.2 0.4 θ 0.6 0.8 1.0
Likelihood (y = 7, J = 10)
Likelihood (y = 14, J = 20)
Likelihood (y = 70, J = 100)
Beta(9,13) Posterior
Beta(16,16) Posterior
Beta(72,40) Posterior

POSTERIOR MEAN

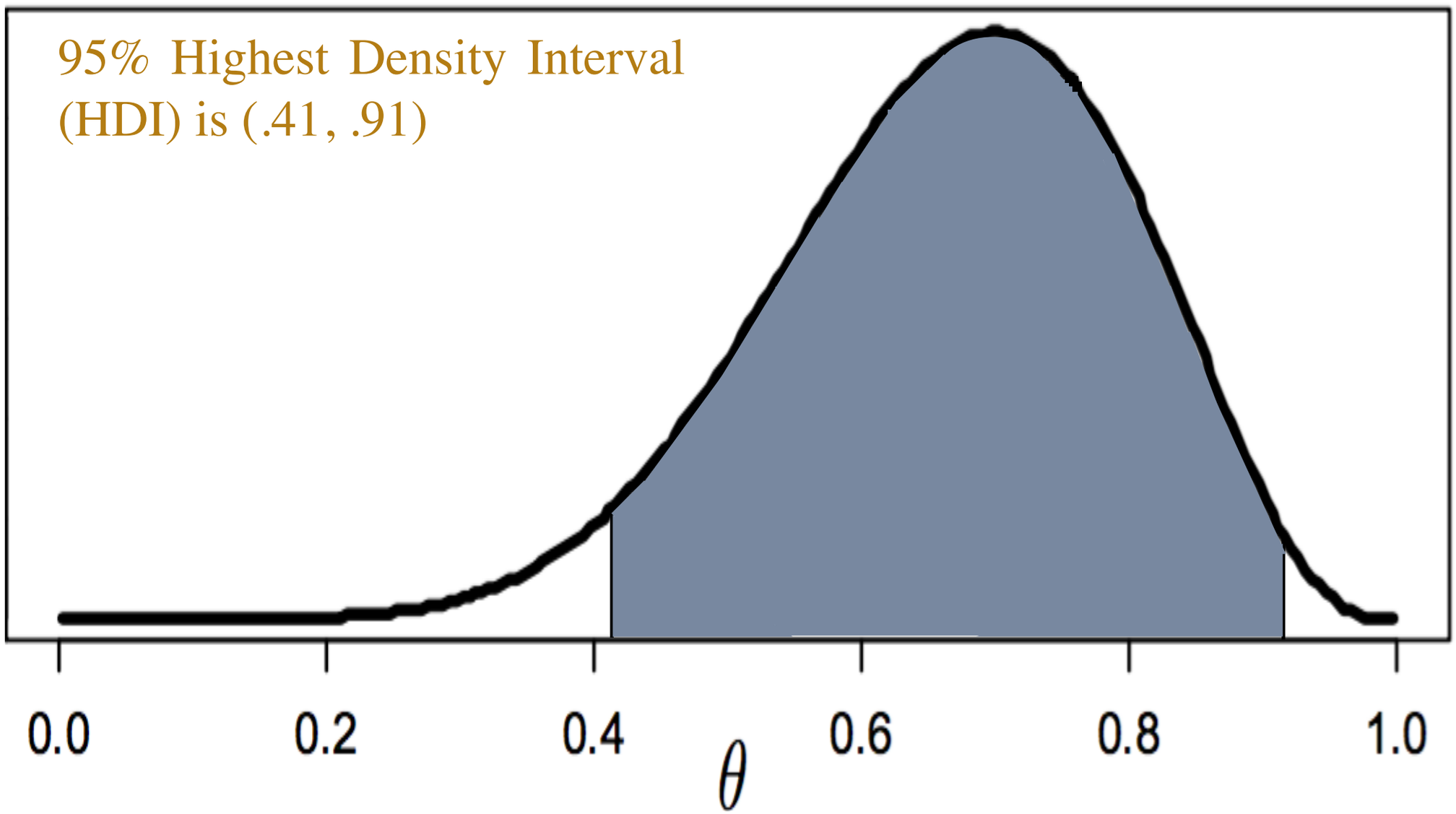
$$p(\theta \mid y, J) = Beta(y + \alpha, J - y + \beta)$$

$$p(\theta \mid y, J) \propto p(y \mid J, \theta)\, p(\theta) \xrightarrow{J \to \infty} p(\theta \mid y, J) \propto p(y \mid J, \theta)$$

$$p(\theta \mid y) \propto p(y \mid \theta)\, p(\theta) \xrightarrow{J \to \infty} p(\theta \mid y) \propto p(y \mid \theta)$$

As J increases, the posterior becomes similar to the likelihood (usually)…

HDI FOR BETA-BINOMIAL



INTERPRETING THE HDI

- The interpretation of the interval is such that probability is ascribed to the **parameter**
- Thus while posterior credibility intervals are often **numerically similar** to frequentist confidence intervals, they have **critically different meanings/interpretations**

"Adopting an explicitly Bayesian approach would resolve a recurring source of confusion for these researchers, letting them say what they mean and mean what they say."

— Jackman (2009)

STRATEGIES FOR OBTAINING POSTERIOR DISTRIBUTIONS

1. **Analytically**. Do the math. This requires the product of the prior and the likelihood taking on a recognizable form and the evaluation of the denominator.

2. **Simulation-based estimation**. Construct a sampling algorithm to simulate or draw from the posterior...*without requiring we know the form of the distribution*. Collect many such draws, which serve to empirically approximate the posterior distribution, and can be used to empirical approximate summary statistics.

Do this via **Markov Chain Monte Carlo (MCMC)** estimation

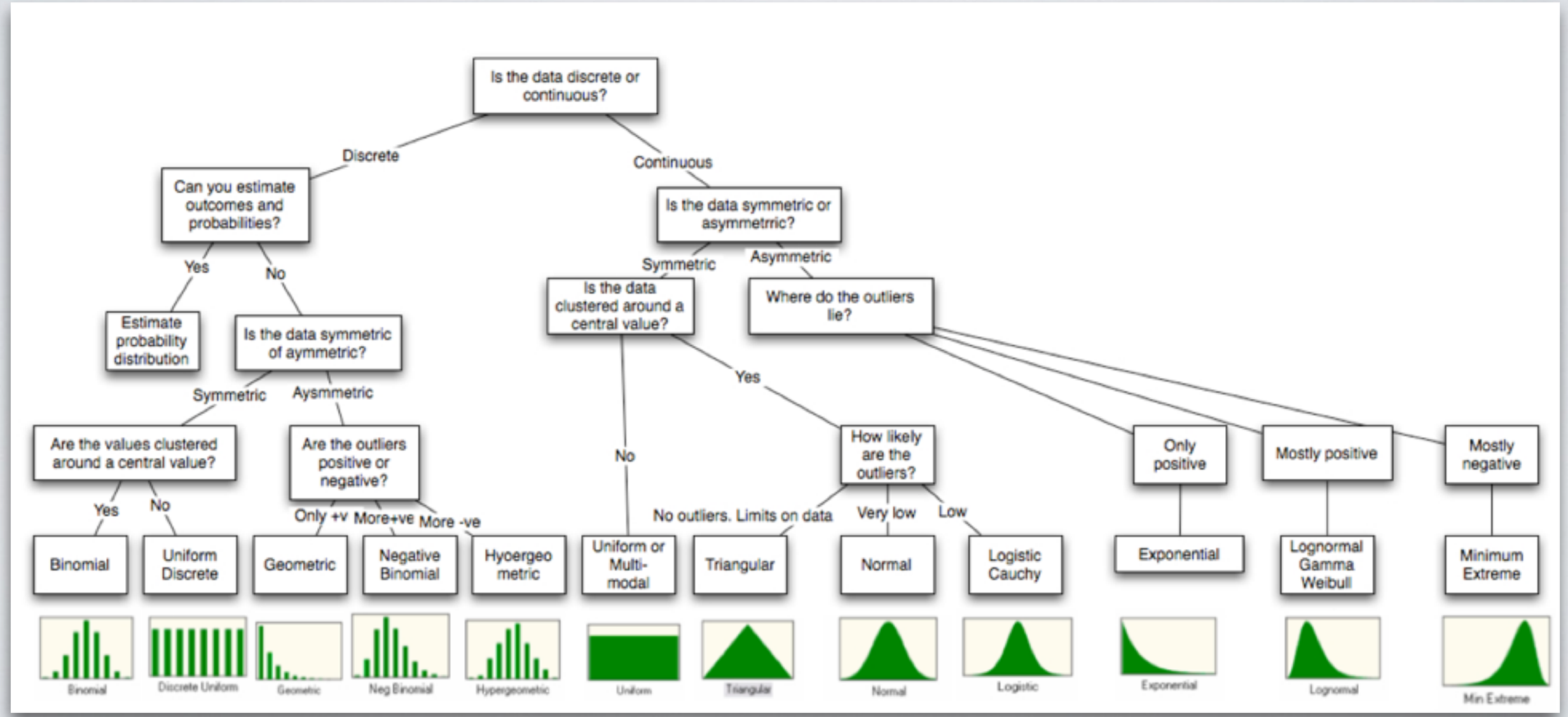
Monte Carlo Principle:

> Anything we want to know about a random variable θ can be learned by sampling many times from $f(\theta)$, the density of θ.
>
> —Jackman (2009)

MCMC: see Roy Levy slides

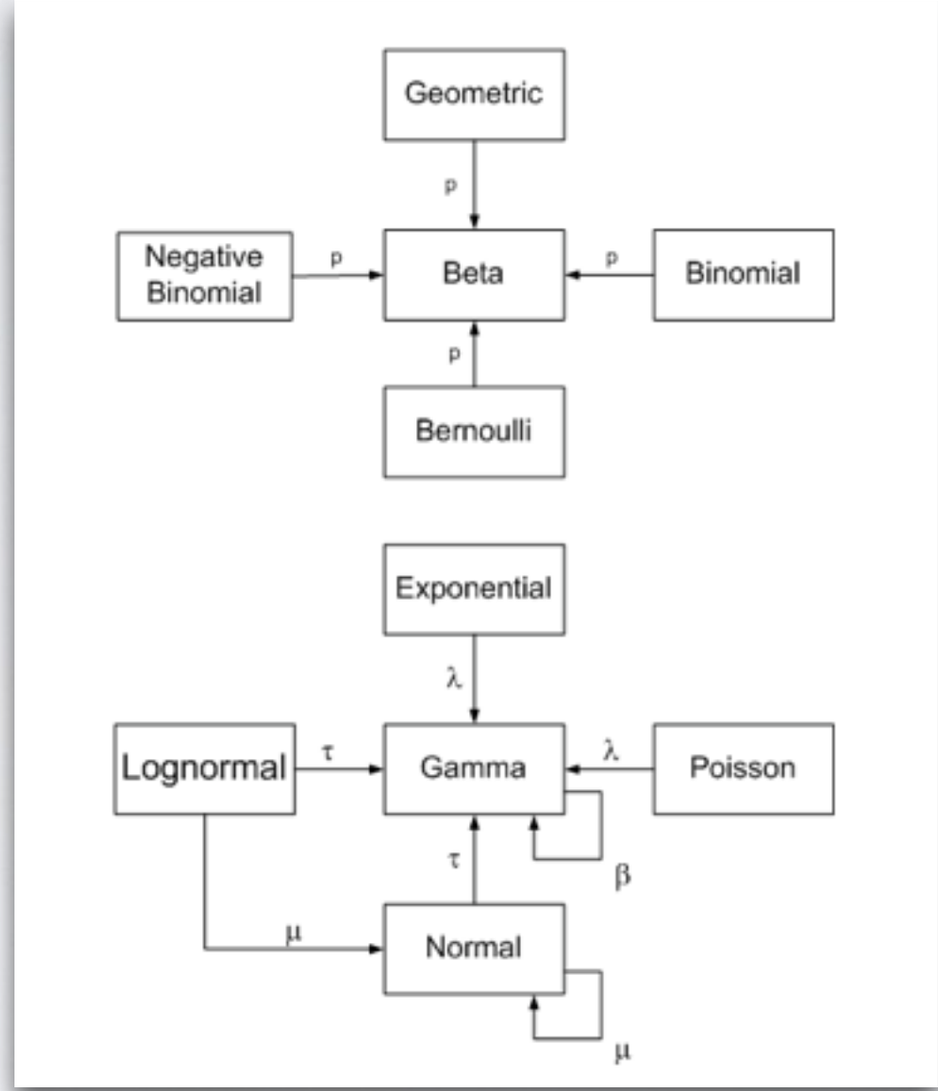
Strategies for specifying prior distribution

DISTRIBUTIONAL CHOICES



COMPUTATIONAL EASE

- How easily can the posterior distribution be obtained ?
- Conjugacy: prior is of a family such that the posterior is of a known family



$p(x|\mu) \sim N, \sigma^2$ *known*, $p(\mu) \sim N \rightarrow p(\mu|x) \sim N$

$p(x \mid \theta) \sim$ *Bern. or Bin.*, $p(\theta) \sim$ *Beta* $\rightarrow p(\theta \mid x) \sim$ *Beta*

DIFFERENT STRATEGIES

- **Strategy 1 : as much ignorance as possible**
  - Prior for the correlation between x and y as $U[-1, 1]$
  - Imprecise prior : $N(0, 1000)$
- **Strategy 2 : use previous data, results**
  - If 19,582 studies have suggested the correlation between X and Y is between .4 and .6 use a prior that puts most of density there
- **Strategy 3 : use theory, knowledge of experts**
  - Often this comes from subject matter experts (SMEs)
  - Worked examples:
    - Almond's (2010) "Linguistic priors" in item response theory, Bayes nets

OBJECTION : DO NOT HAVE PRIOR BELIEFS

"I don't have prior beliefs."

- For the correlation (stdized slope, etc.), do you really believe that a value of .9 is just as likely as -.9?
- When an estimate of a variance <0, do you really believe that?
- Proportion of such patients with breast cancer = .008
- **Don't have prior beliefs → you haven't done your homework**

OBJECTION : DO NOT WANT SUBJECTIVITY

"I may have prior beliefs, but I don't want them to cloud the analysis. The analysis should objective."

Typical applications of regression assume y | x ~ N

- Why use a N, rather than t for the likelihood?
- Why do a linear model, in the likelihood? With those parameters, not others?
- How do you determine outliers? Decide what to do with them?
- Your choice of α? n? $\mathbf{x}$?
- Subjective ≠ Arbitrary

Model checking

EXAMPLE

- Pedometer readings from students during a day of school
  - 16 girls, 13 boys
  - Measured in 100s of step
- 2-group normal model setup
- For group 1 (girls) $x_i \sim N(\mu_1, \sigma_1^2)$
- For group 2 (boys) $x_i \sim N(\mu_2, \sigma_2^2)$

	Girls	Boys
	49.93	63.19
	61.68	75.72
	60.41	42.39
	70.63	99.05
	53.63	50.21
	29.05	62.89
	41.60	32.20
	51.80	60.45
	72.34	110.93
	43.12	52.40
	47.91	84.12
	50.58	103.52
	53.56	90.64
	35.92	
	56.96	
	24.03	
Mean	50.20	71.36
Standard Deviation	13.27	24.76

INDEPENDENT SAMPLES T-TEST

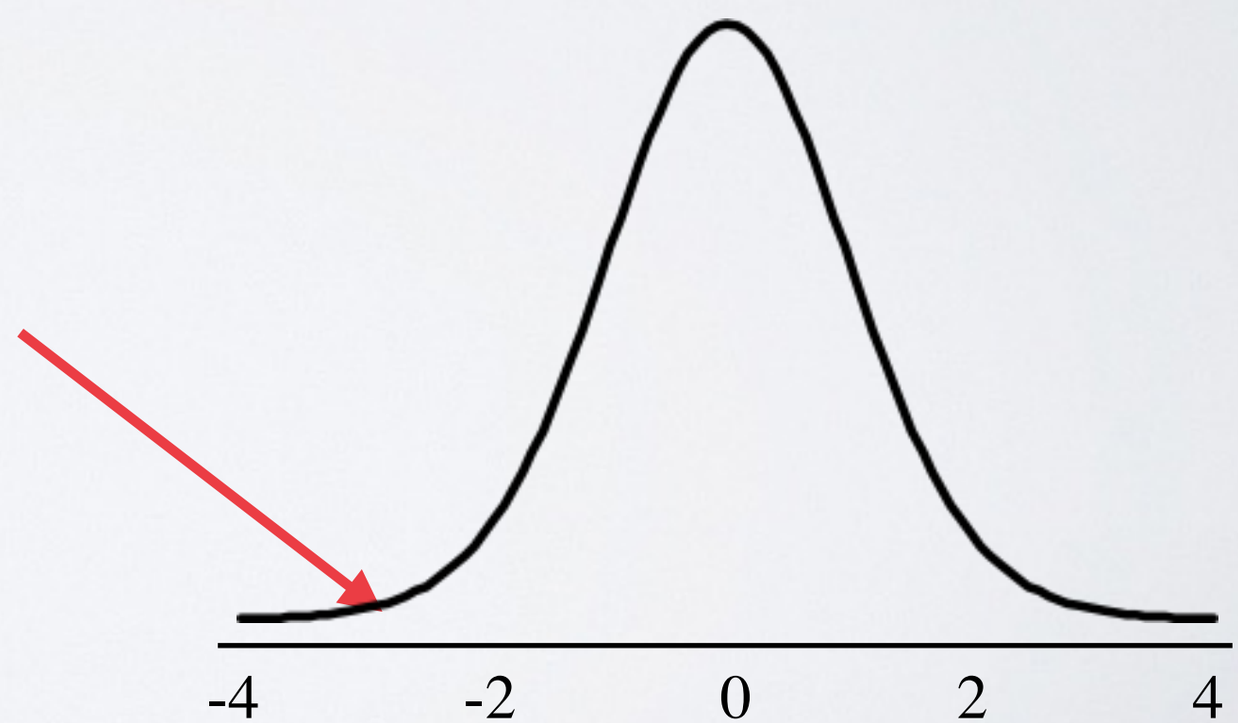
Test of difference in population means, $\mu_D = \mu_1 - \mu_2$

H_0: $\mu_D = 0$

H_A: $\mu_D \neq 0$

t distribution with 27 df

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - \mu_D}{s_{\bar{X}_1 - \bar{X}_2}} = -2.94$$



INDEPENDENT SAMPLES T-TEST

Test of difference in population means, $\mu_D = \mu_1 - \mu_2$

H_0: $\mu_D = 0$

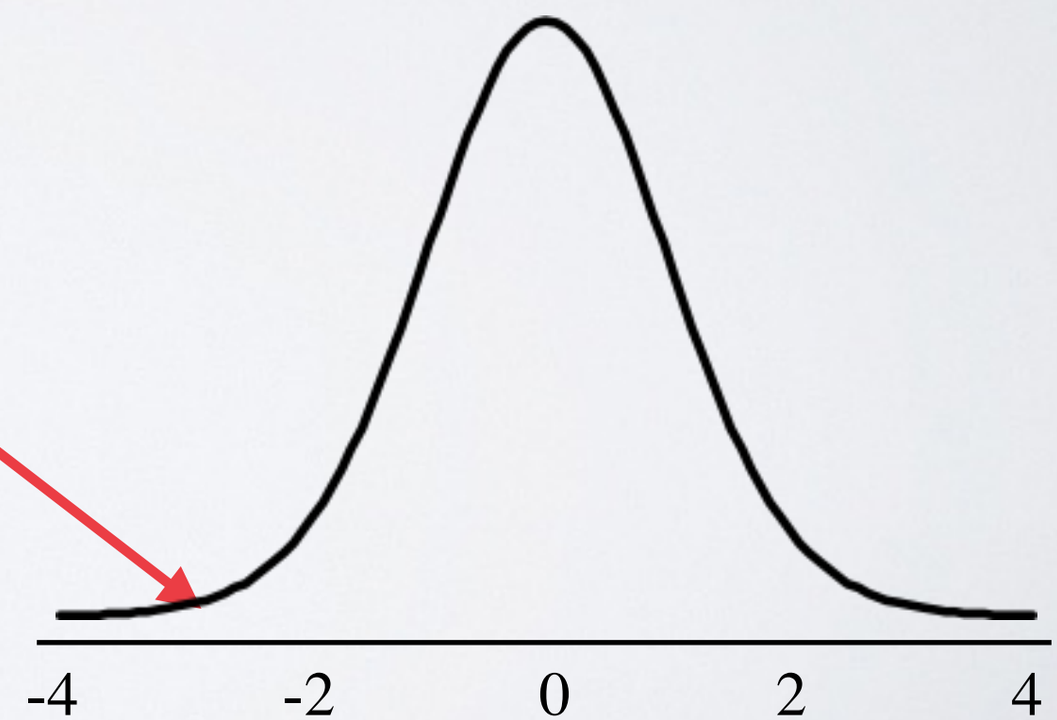
H_A: $\mu_D \neq 0$

Distribution reflects implications of H_0: $\mu_D = 0$, rooted in $p(x|\theta)$

t distribution with 27 df

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - \mu_D}{s_{\bar{X}_1 - \bar{X}_2}} = -2.94$$

Locate the observed data relative to this reference distribution



A bayesian approach

BAYESIAN MODEL FOR EXAMPLE

- 2-group normal model setup
- For group 1 (girls) $x_i \sim N(\mu_1, \sigma_1^2)$
- For group 2 (boys) $x_i \sim N(\mu_2, \sigma_2^2)$
- **Evaluate a model where $\mu_1 = \mu_2 = \mu$ and $\sigma_1^2 = \sigma_2^2 = \sigma^2$**
- $\mu \sim N(50, 1000)$
- $\sigma^2 \sim Inv\text{-}Gamma(1, 1)$

BAYESIAN MODEL FOR EXAMPLE

```
for(i in 1:n.1){
   x.1[i] ~ dnorm(mu, tau)
} # closes loop over subjects

for(i in 1:n.2){
   x.2[i] ~ dnorm(mu, tau)
} # closes loop over subjects

mu ~ dnorm(mu.mu, tau.mu)
```

POSTERIOR PREDICTIVE MODEL CHECKING VIA TEST STATISTICS

How well does the model recover or account for this characteristic or feature of the data?

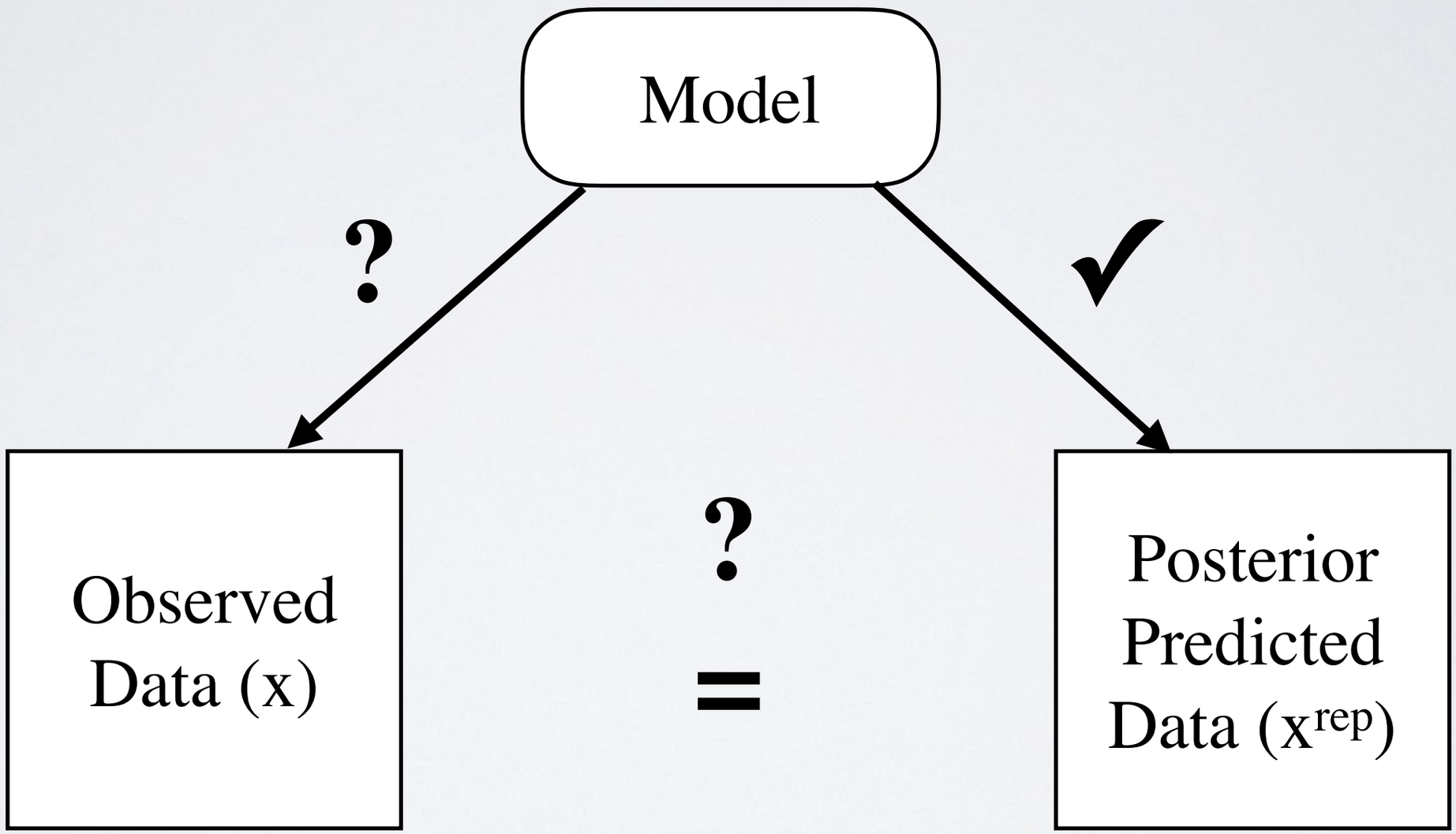
1. What is the characteristic or feature of the data of interest?

   - Realised test statistic T(x) ($\mu 1, \mu 2, (\mu 1\text{-}\mu 2), \ldots$)

2. Given the model, what would be expected for this characteristic or feature of the data?

3. How does the characteristic or feature of the data compare to what the model implies the characteristic or feature ought to be?

COMPARING OBSERVED AND POSTERIOR PREDICTIVE DATA



POSTERIOR PREDICTIVE DISTRIBUTION FOR THE TEST STATISTICS

How does the characteristic or feature of the data compare to what the model implies the characteristic or feature ought to be?

- The posterior predictive distribution of the test statistic serves as a ***reference distribution*** against which the observed value is compared

- Compare the realized value $T(\mathbf{x})$ to the posterior predictive distribution $T(\mathbf{x}^{rep(1)}),\ldots,T(\mathbf{x}^{rep(T)})$

Model comparison

BAYES FACTOR

$$\frac{p(M_1 \mid x)}{p(M_2 \mid x)} = \frac{p(x \mid M_1)}{p(x \mid M_2)} \times \frac{p(M_1)}{p(M_2)}$$

Posterior odds

Bayes factor

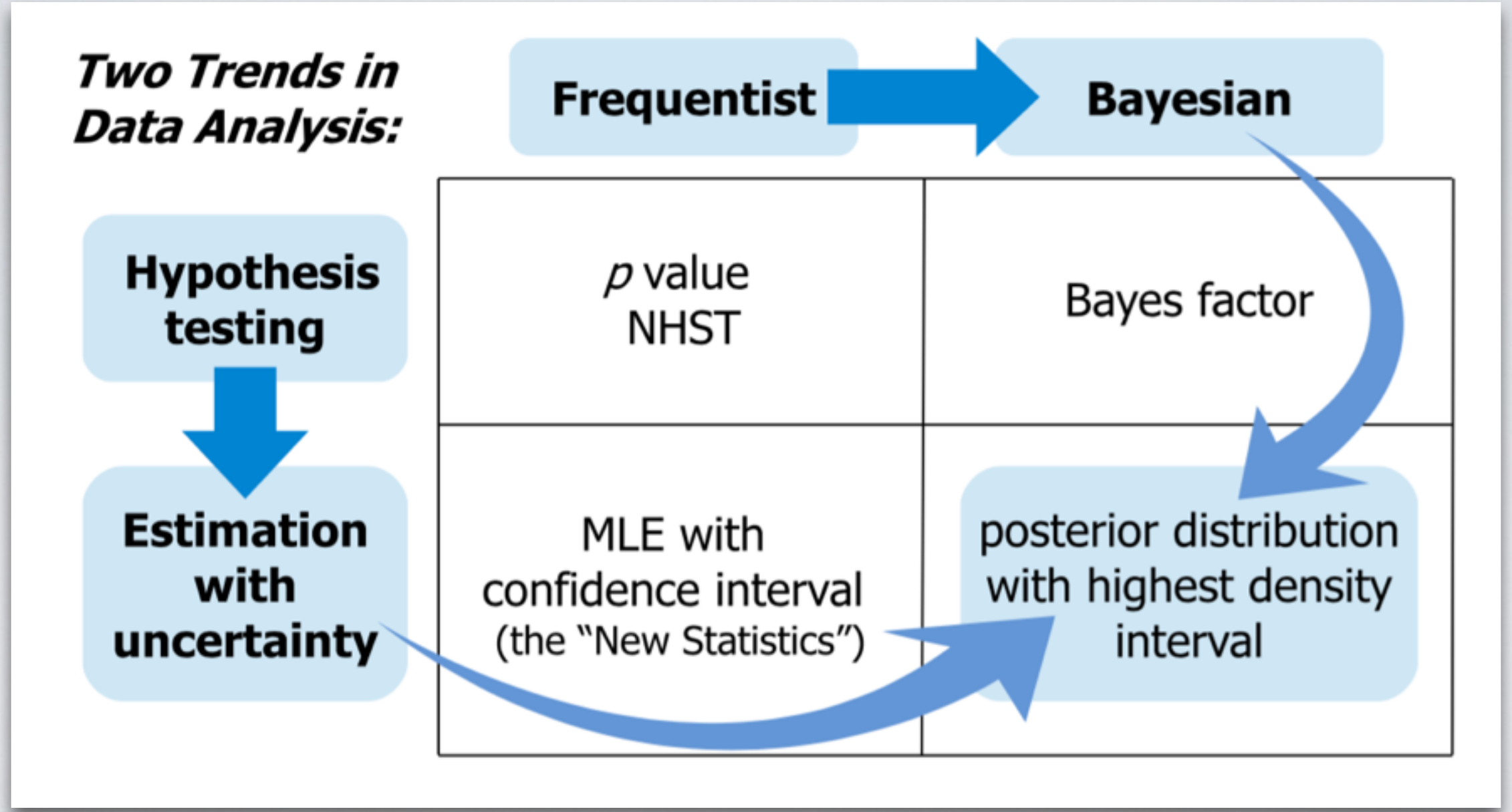
Prior odds

BAYES FACTOR

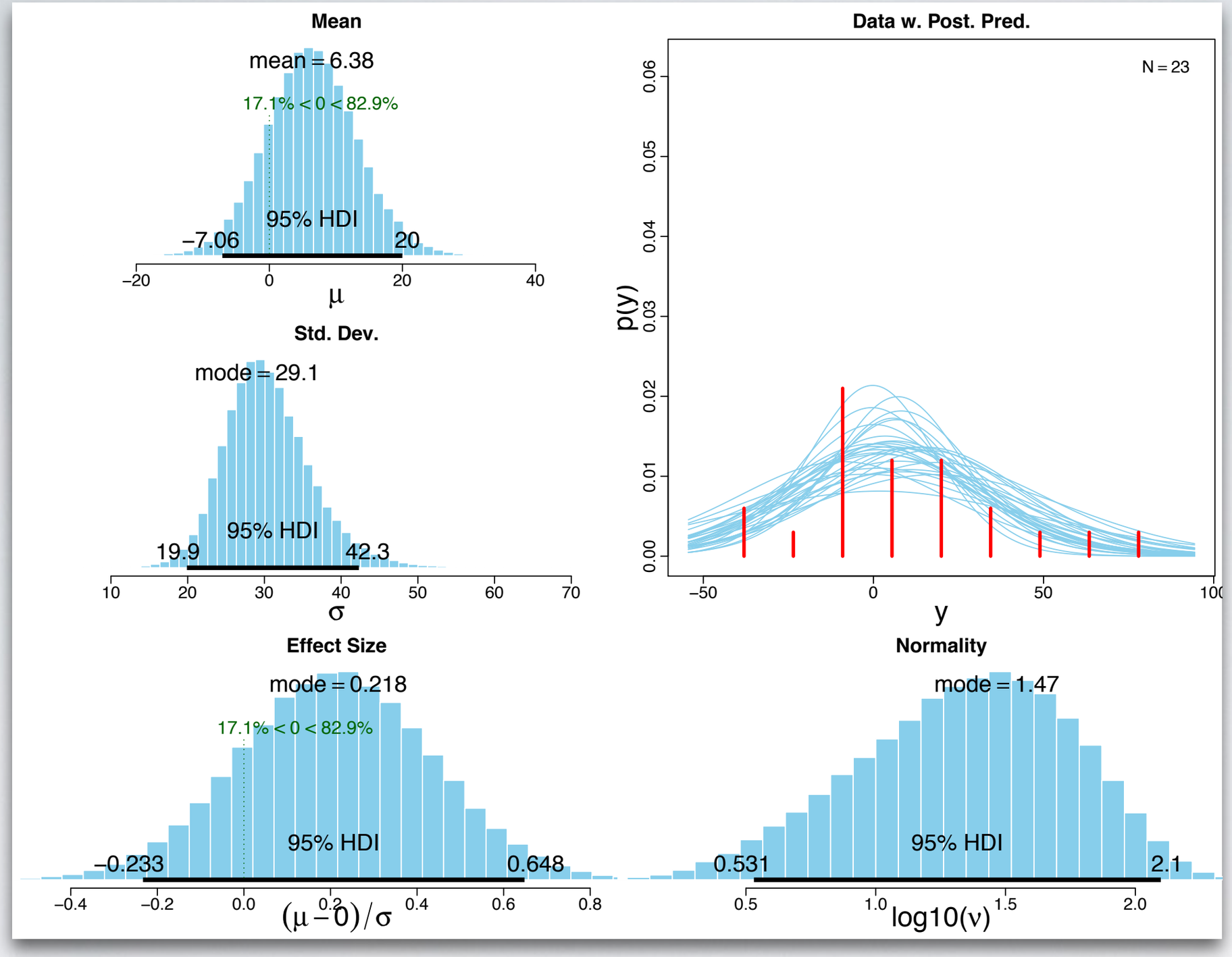
$$\frac{p(x \mid M_1)}{p(x \mid M_2)} = \frac{\int p(x \mid \theta_1) p(\theta_1) d\theta_1}{\int p(x \mid \theta_2) p(\theta_2) d\theta_2}$$

- Each term is essentially the marginal distribution of the data under the respective model (where "model" means prior distribution and likelihood)
- Specializes to a likelihood ratio in some cases

TWO TRENDS IN DATA ANALYSIS



Kruschke & Liddell, 2015



Package "BEST" (Bayesian Estimation Supersedes the T-test, Kruschke & Meredith, 2015)

Summary

A ***Bayesianly justifiable*** analysis is one that

> *"treats known values as observed values of random variables, treats unknown values as unobserved random variables, and calculates the conditional distribution of unknowns given knowns and model specifications using Bayes' theorem."*

— Rubin (1984)

3 STEP APPROACH TO BAYESIAN MODELING

1. **Set up the full probability model**: the joint distribution of all entities, including observables (x) and unobservables (θ) in accordance with all that is known about the problem

$$p(x, \theta) = p(x \mid \theta)\, p(\theta)$$

2. Condition on the observed data (x), calculate the conditional probability distribution for the unobservable entities (θ) of interest given the observed data: obtain the **posterior distribution** as

$$p(\theta \mid x) = \frac{p(x \mid \theta) p(\theta)}{p(x)}$$

3. Examine fit, tenability/sensitivity of assumptions, reasonable, summarize results, etc.

"These days the statistician is often asked such questions as "Are you a Bayesian?" "Are you a frequentist?" "Are you a data analyst?" "Are you a designer of experiments?" I will argue that the appropriate answer to all these questions can be (and preferably should be) "yes"...

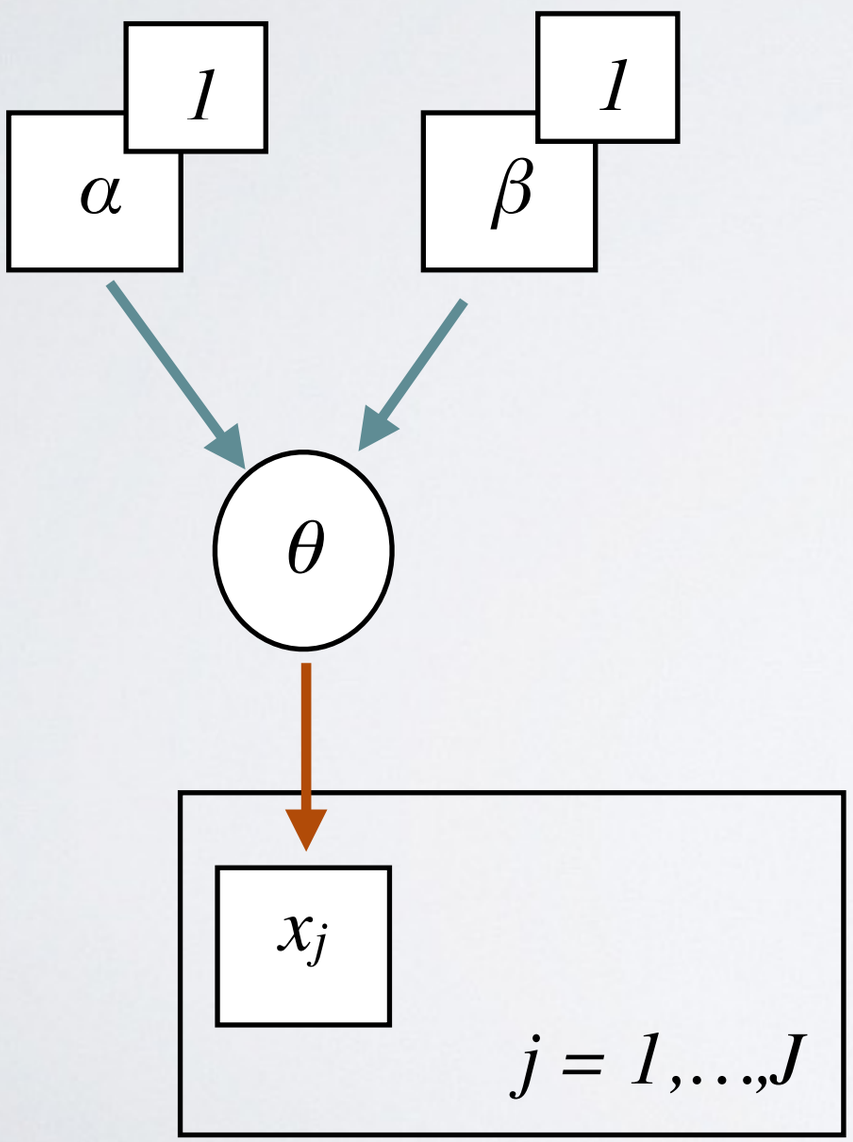
— Box (1983)

The end

(+Appendixes)

GRAPH, EQUATIONS, CODE

$x_1, \ldots, x_J$ is the result of a series of J Bernoulli trials with probability θ



$$\theta \sim \text{Beta}(1, 1)$$

$$x_j \mid \theta \sim \text{Bern}(\theta)$$

$$j = 1, \ldots, J$$

```
theta ~ dbeta(1,1)

for(j in 1:J){
dx[j] ~ dbern(theta)
}
```

Regression

SIMPLE REGRESSION MODEL

- Two variables, x and y for each of n subjects

$$\mathbf{x} = (x_1, x_2, x_3, \ldots, x_n)$$

$$\mathbf{y} = (y_1, y_2, y_3, \ldots, y_n)$$

- The (conditional) probability of the data is given by:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

$$\varepsilon_i \text{ independent}, \sim N(0, \sigma_\varepsilon^2)$$

$$y_i \mid x_i, \beta_0, \beta_1, \sigma_\varepsilon^2 \sim N(\beta_0 + \beta_1 x_i, \sigma_\varepsilon^2)$$

CONDITIONAL PROBABILITY OF THE DATA

$$p(\beta_0, \beta_1, \sigma_\varepsilon^2 \mid \boldsymbol{y}, \boldsymbol{x}) \propto p(\boldsymbol{y} \mid \beta_0, \beta_1, \sigma_\varepsilon^2, \boldsymbol{x})\, p(\beta_0, \beta_1, \sigma_\varepsilon^2)$$

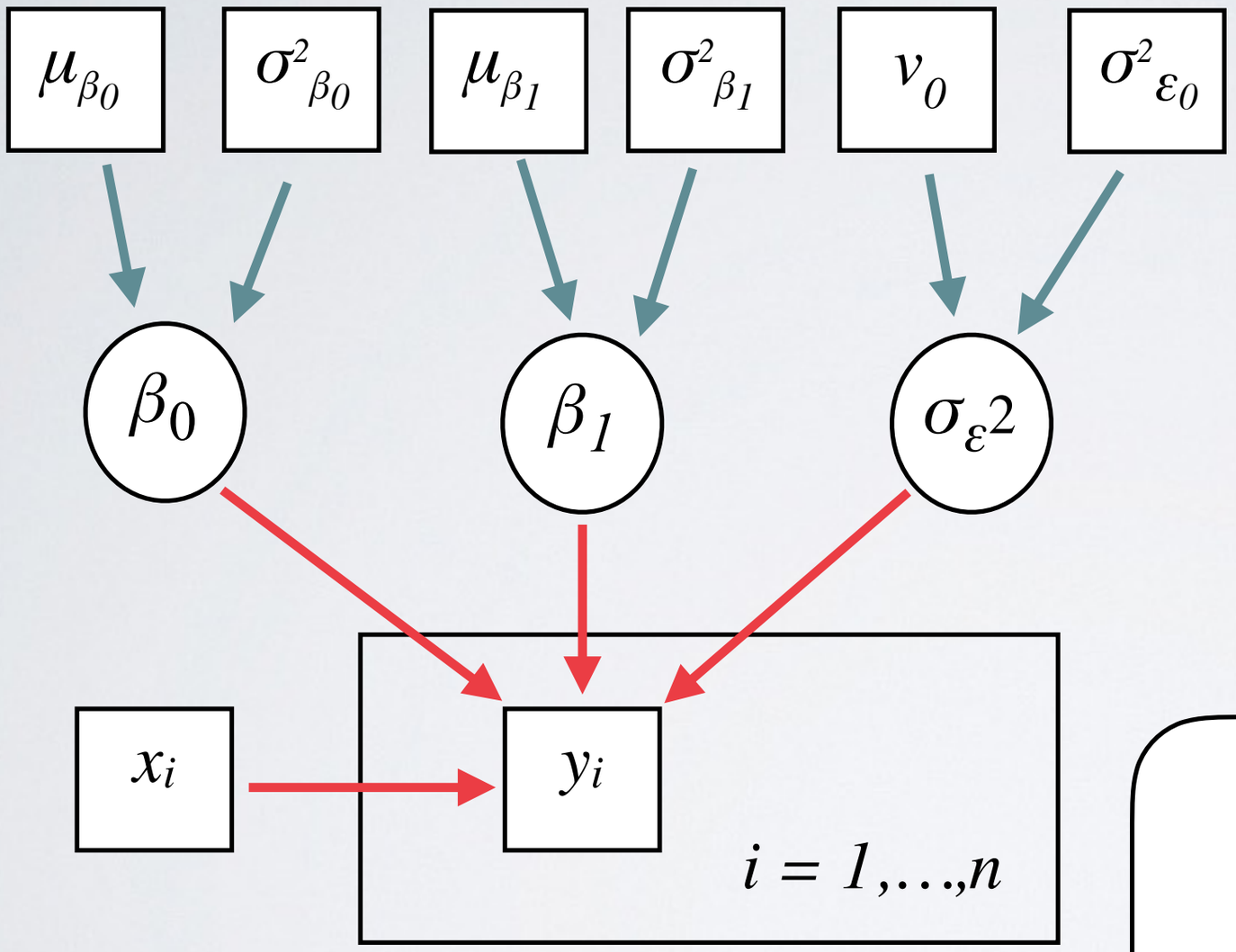
Assuming independence of subjects

$$p(\boldsymbol{y} \mid \beta_0, \beta_1, \sigma_\varepsilon^2, \boldsymbol{x}) = \Pi_i\, p(y_i \mid \beta_0, \beta_1, \sigma_\varepsilon^2, \mathbf{x}_i)$$

Assuming conditional normality

$$y_i \mid \beta_0, \beta_1, \sigma_\varepsilon^2, \mathbf{x}_i \sim N(\beta_0 + \beta_1 x_i, \sigma_\varepsilon^2)$$

GRAPHICAL MODEL




```
theta ~ dbeta(1,1

for(j in 1:J){
dx[j] ~ dbern(theta)
}
```

$$p(\beta_0, \beta_1, \sigma_\varepsilon^2) = p(\beta_0)\, p(\beta_1)\, p(\sigma_\varepsilon^2)$$

$$\beta_0 \sim N(\mu_{\beta_0}, \sigma^2_{\beta_0})$$

$$\beta_1 \sim N(\mu_{\beta_1}, \sigma^2_{\beta_1})$$

$$\sigma_\varepsilon^2 \sim \text{Inv-Gamma}(\nu_0 / 2, \nu_0 \sigma^2_{\varepsilon_0}/2)$$

BAYESIAN INFORMATION CRITERION (SCHWARZ, 1978)

$$BIC = -2\ell(\hat{\theta} \mid x) + p + n$$

- Where $l(\boldsymbol{\theta} \mid \boldsymbol{x})$ is the maximized log-likelihood, p is the number of parameters, n is sample size
- Let $S = (.5)(BIC_{\text{Model 1}} - BIC_{\text{Model 2}})$
- BF $\approx e^{-S}$
- More commonly, select model with smallest *BIC*
- Similar to application of *AIC*, other information criteria

Frequentist modeling

FREQUENTIST MODEL FOR EXAMPLE

- 2-group normal model setup
- For group 1 (girls) $x_i \sim N(50, 1000)$
- For group 2 (boys) $x_i \sim N(50, 1000)$
- Evaluate a model where $\mu_1 = \mu_2 = \mu$ and $\sigma_1^2 = \sigma_2^2 = \sigma^2$
- Hypothesis testing
  - H_0: Model-implied distribution of data $p(x \mid \theta)$ is correct
  - H_A: Model-implied distribution of data $p(x \mid \theta)$ is not correct
  - Test statistic from likelihood theory asymptotically $\sim \chi^2$

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  - H_0: Model-implied distribution of data $p(x \mid \theta)$ is correct
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  - Test statistic from likelihood theory asymptotically $\sim \chi^2$

```r
#------------------------- Bayesian analysis - induction

priors <- list(muM = 0, muSD = 100)

x <- DF$VAS[DF$Session=="Post-induction"] - DF$VAS[DF$Session=="Baseline"]

induc <- BESTmcmc(x, priors = priors, parallel=FALSE)

plot(induc)
plotAll(induc )

summary(induc)
```

```
> summary(induc)
          mean median   mode HDI%  HDIlo HDIup compVal %>compVal
mu       6.398  6.356  6.257   95 -7.260 19.65       0      83.2
sigma   30.482 29.994 28.736   95 19.877 42.12
nu      32.650 23.542  7.651   95  0.980 92.31
log10nu  1.340  1.372  1.455   95  0.515  2.08
effSz    0.212  0.213  0.209   95 -0.230  0.65       0      83.2
```